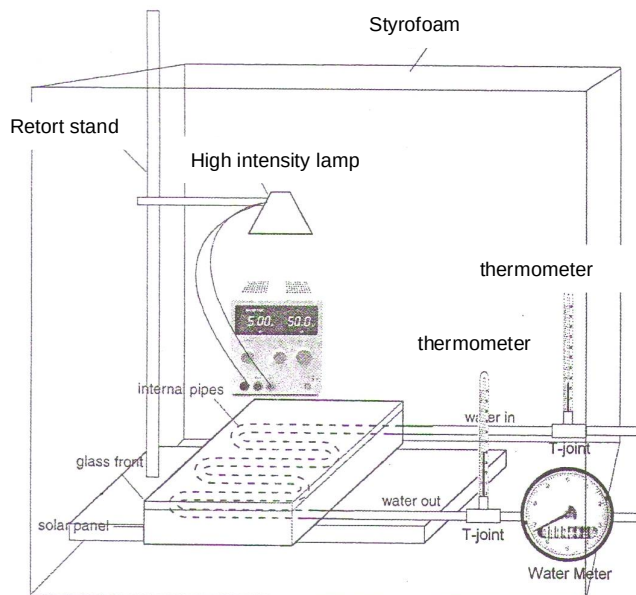


7

Diagram:



Aim: To investigate the efficiency of a solar panel under different illumination and flow rate conditions.

Apparatus: High intensity lamp
Variable power supply
Intensity meter of Infra Red
Solar Panel
Thermometers
Styrofoam box
Retort stand
Water flow meter
Stop watch
Electronic balance
Measuring cylinder

Independent variable: efficiency of solar panel

The efficiency of solar panel can be calculated by

$$Q = mc\Delta\theta$$

$$P_{Out} = \frac{dm}{dt} c\Delta\theta$$

$$P_{In} = I_{intensity} A_{area\ of\ solar\ panel}$$

$$\eta_{Eff} = \frac{P_{Out}}{P_{In}} \times 100$$

Dependent variable: *Intensity and flow rate of water*

Control variables:

- intensity meter at constant height above panel
- Inflow water kept at constant temperature.

Using the same variable power supply.

Procedure:

- Set up apparatus as shown in diagram.
- Place solar panel setup in Styrofoam box to keep out heat from surroundings. .
- Hold the high intensity lamp at a fixed distance above the solar panel using a retort stand.
- Place one thermometer each at the inlet and outlet T-joints.
- Keeping the flow rate constant with the inlet tap at a fixed setting, the intensity of the high intensity lamp is varied by adjusting the output voltage of the variable power supply.
- With each intensity setting, the intensity of the infra-red radiation arriving at the solar panel is measured by the intensity-meter.
- The intensity meter should be placed at the same vertical height above the table as that of the pipes.
- Measure 6 different intensity settings of infra-red radiation for 6 different flow rate.
- Measure the difference between inlet and outlet temperature for each flow rate.

Calculation for efficiency

$$Q = mc\Delta\theta$$

$$P_{Out} = \frac{dm}{dt} c\Delta\theta$$

$$P_{In} = I_{intensity} A_{area\ of\ solar\ panel}$$

$$\eta_{Eff} = \frac{P_{Out}}{P_{In}} \times 100$$

Hypothesis:

The efficiency η_{Eff} may be related to average intensity by the equation

$$\eta_{Eff} = kI^Z$$

Plot $\log \eta_{Eff}$ vs $\log I$

Tabulation:

Average Incident Intensity/ $W\ m^{-2}$	Change in temperature/ $^{\circ}C$	Average power output P_{out}/W	Average power incident P_{in}/W	Efficiency η	$\lg I$	$\lg \eta$

- Plot a graph of $\lg \eta$ against $\lg I$. Determine the value of Z from the gradient of the graph since $\lg \eta = Z \lg I + \lg k$.

Precaution: Enclose the experimental setup in a Styrofoam box to minimize ambient heat.

Safety: Avoid shorting the leads of the variable power supply.
 Avoid touching the leads of the variable power supply to prevent electric shock.

Note:

- (i) Spent too much time drawing neat diagrams
- (ii) Intensity meter ignored.
- (iii) Encourage clear fully labelled diagram to reduce unnecessary explanation.
- (iv) Give clear variables investigated and constant variables
- (v) Refrain from attaching voltmeter and ammeter to solar panel.
- (vi) Read question carefully!